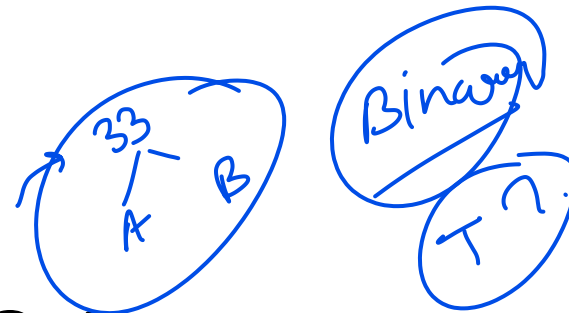


Lec_34



Practice Questions on Binary Search

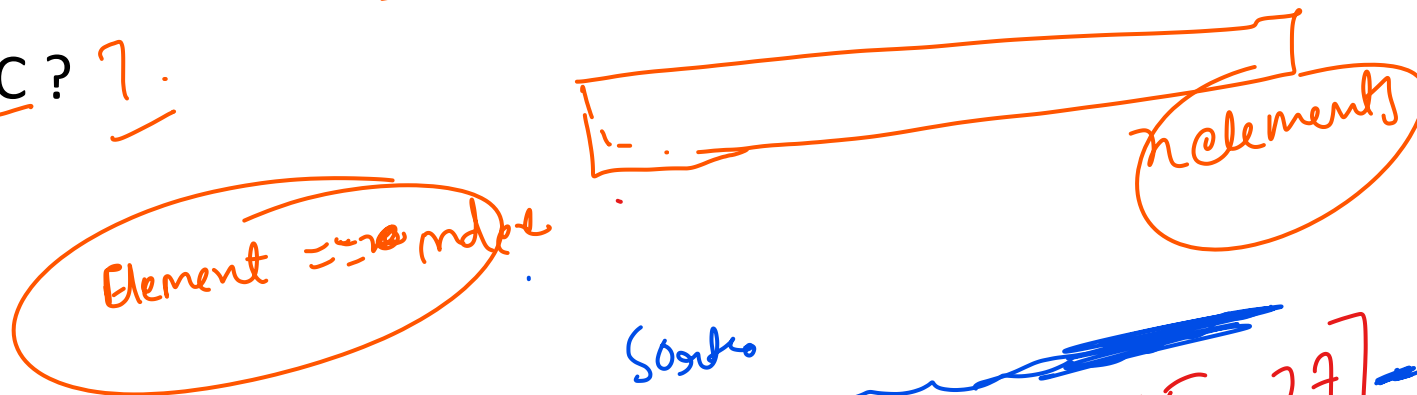
↓ gate?

1.) i/p: A sorted array of n -distinct element
 o/p: find any element $a[i]$, such that $a[i]=i$;

$$TC = O(\log n)$$

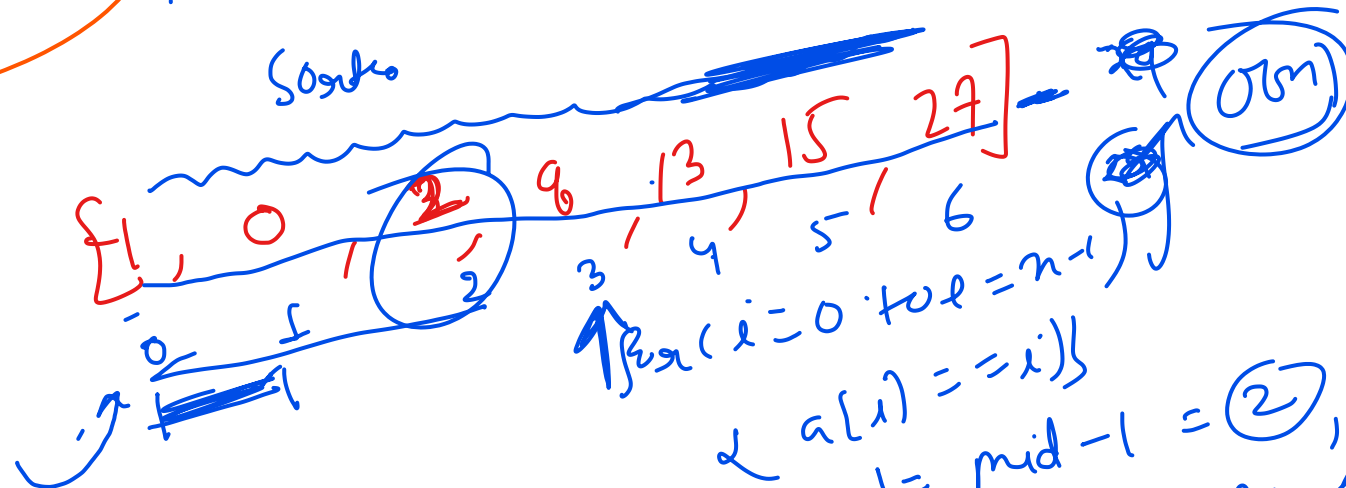
$$SC = O(1)$$

• TC and SC ?



$a[i] = i$

Ex:



Run:

$$l = 0, r = 6$$

$$mid = \frac{l+r}{2} = \frac{0+6}{2} = 3$$

$$a[mid] = 9$$

$$a[mid] > mid$$

$$a[l] = a[0] = 1$$

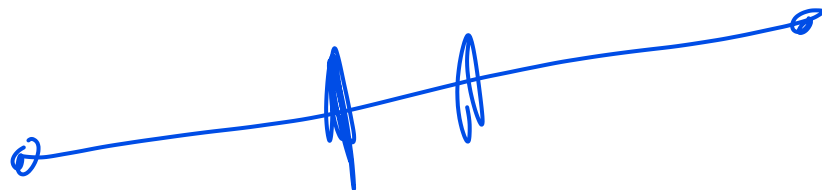
$$r = mid - 1 = 2$$

$$\frac{l+r}{2} = \frac{0+2}{2} = 1$$

$\textcircled{11} \quad \text{mid} = 1, a(\text{mid}) = 0$
 $a(\text{mid}) < \text{mid}$

~~$l = 2$~~
 ~~$j = 2$~~

$\textcircled{3} \quad m = \frac{l+j}{2} = \frac{1+1}{2} = 1$



algo:

```
fun(a, i, j)
{
  if (i == j) {
    if (a[i] == i) return i;
    else return -1;
  }
```

```
  else {
    mid =  $\frac{i+j}{2}$ 
    if (a[mid] == mid) {
      return mid;
    }
    if (a[mid] > mid) {
      fun(a, i, mid-1);
    }
    else {
      geten
      fun(a, mid+1, j);
    }
  }
```

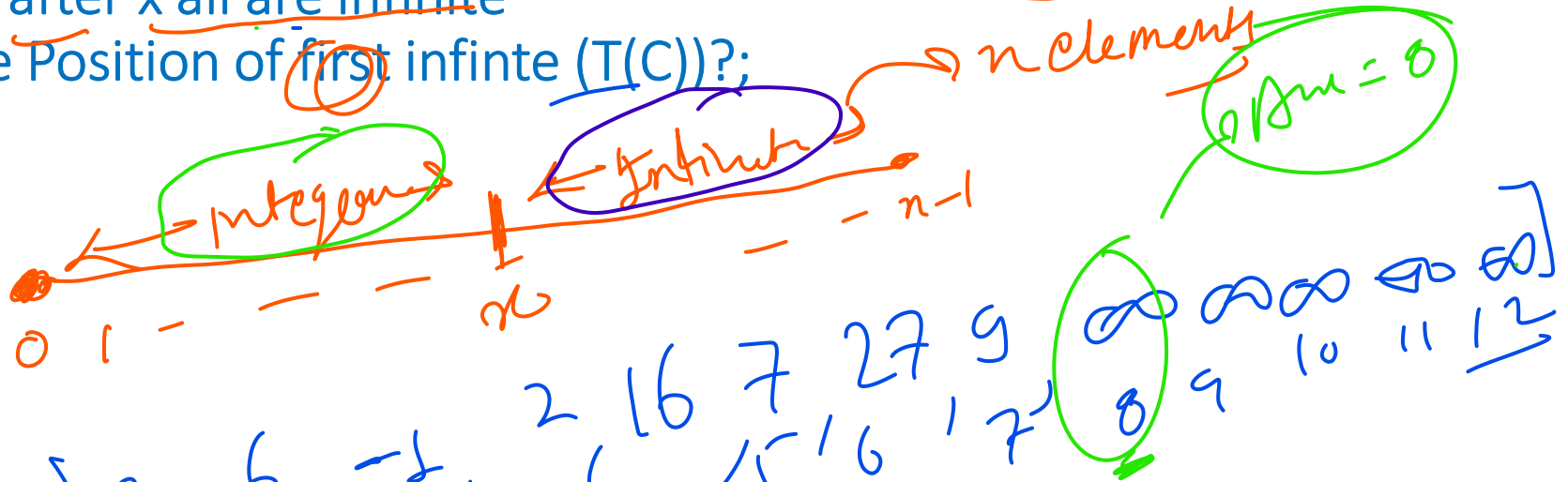
$O(\log n)$
 $O(\log n)$

2.) i/p: an array of n-element, in which until some position x all are integers and after x all are infinite

o/p: find the Position of first infinite (T(C))?

TC: 1

Ex:



→ Linear search = $O(n)$

```
for (i = 0; i < n; i++)
    if (a[i] == ∞)
        return i
```

#

$[3, 5, 7, 17, 1, 1, 0, -13, \infty, \infty, \infty, \infty]$
 $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$

1)

$$l=0, j=12$$

$$mid = \frac{l+j}{2} = \frac{12}{2} = 6$$

2)

$$l = mid + 1 = 6 + 1 = 7, j = 12$$

$$mid = \frac{7+12}{2} = 9$$

$a[mid] = \infty$

3)

$$l = 7, j = 8$$

$$mid = \frac{7+8}{2} = 7.5 \rightarrow 7$$

$a[mid] = -13$

gefunden $(mid + 1)$

$TC: O(\log n)$
 $SC: O(1)$

Logn

→ offen



Basic thing:-

$$2^x = 2^{x \log n}$$

A $l=0$, $l=2+1=2$, $l=2^x$, $l=4$, $l=6$, $l=8$, $l=10$, $l=12$, $l=14$, $l=16$, $l=18$, $l=20$, $l=22$, $l=24$, $l=26$, $l=28$, $l=30$, $l=32$, $l=34$, $l=36$, $l=38$, $l=40$, $l=42$, $l=44$, $l=46$, $l=48$, $l=50$, $l=52$, $l=54$, $l=56$, $l=58$, $l=60$, $l=62$, $l=64$, $l=66$, $l=68$, $l=70$, $l=72$, $l=74$, $l=76$, $l=78$, $l=80$, $l=82$, $l=84$, $l=86$, $l=88$, $l=90$, $l=92$, $l=94$, $l=96$, $l=98$, $l=100$, $l=102$, $l=104$, $l=106$, $l=108$, $l=110$, $l=112$, $l=114$, $l=116$, $l=118$, $l=120$, $l=122$, $l=124$, $l=126$, $l=128$, $l=130$, $l=132$, $l=134$, $l=136$, $l=138$, $l=140$, $l=142$, $l=144$, $l=146$, $l=148$, $l=150$, $l=152$, $l=154$, $l=156$, $l=158$, $l=160$, $l=162$, $l=164$, $l=166$, $l=168$, $l=170$, $l=172$, $l=174$, $l=176$, $l=178$, $l=180$, $l=182$, $l=184$, $l=186$, $l=188$, $l=190$, $l=192$, $l=194$, $l=196$, $l=198$, $l=200$, $l=202$, $l=204$, $l=206$, $l=208$, $l=210$, $l=212$, $l=214$, $l=216$, $l=218$, $l=220$, $l=222$, $l=224$, $l=226$, $l=228$, $l=230$, $l=232$, $l=234$, $l=236$, $l=238$, $l=240$, $l=242$, $l=244$, $l=246$, $l=248$, $l=250$, $l=252$, $l=254$, $l=256$, $l=258$, $l=260$, $l=262$, $l=264$, $l=266$, $l=268$, $l=270$, $l=272$, $l=274$, $l=276$, $l=278$, $l=280$, $l=282$, $l=284$, $l=286$, $l=288$, $l=290$, $l=292$, $l=294$, $l=296$, $l=298$, $l=300$, $l=302$, $l=304$, $l=306$, $l=308$, $l=310$, $l=312$, $l=314$, $l=316$, $l=318$, $l=320$, $l=322$, $l=324$, $l=326$, $l=328$, $l=330$, $l=332$, $l=334$, $l=336$, $l=338$, $l=340$, $l=342$, $l=344$, $l=346$, $l=348$, $l=350$, $l=352$, $l=354$, $l=356$, $l=358$, $l=360$, $l=362$, $l=364$, $l=366$, $l=368$, $l=370$, $l=372$, $l=374$, $l=376$, $l=378$, $l=380$, $l=382$, $l=384$, $l=386$, $l=388$, $l=390$, $l=392$, $l=394$, $l=396$, $l=398$, $l=400$, $l=402$, $l=404$, $l=406$, $l=408$, $l=410$, $l=412$, $l=414$, $l=416$, $l=418$, $l=420$, $l=422$, $l=424$, $l=426$, $l=428$, $l=430$, $l=432$, $l=434$, $l=436$, $l=438$, $l=440$, $l=442$, $l=444$, $l=446$, $l=448$, $l=450$, $l=452$, $l=454$, $l=456$, $l=458$, $l=460$, $l=462$, $l=464$, $l=466$, $l=468$, $l=470$, $l=472$, $l=474$, $l=476$, $l=478$, $l=480$, $l=482$, $l=484$, $l=486$, $l=488$, $l=490$, $l=492$, $l=494$, $l=496$, $l=498$, $l=500$, $l=502$, $l=504$, $l=506$, $l=508$, $l=510$, $l=512$, $l=514$, $l=516$, $l=518$, $l=520$, $l=522$, $l=524$, $l=526$, $l=528$, $l=530$, $l=532$, $l=534$, $l=536$, $l=538$, $l=540$, $l=542$, $l=544$, $l=546$, $l=548$, $l=550$, $l=552$, $l=554$, $l=556$, $l=558$, $l=560$, $l=562$, $l=564$, $l=566$, $l=568$, $l=570$, $l=572$, $l=574$, $l=576$, $l=578$, $l=580$, $l=582$, $l=584$, $l=586$, $l=588$, $l=590$, $l=592$, $l=594$, $l=596$, $l=598$, $l=600$, $l=602$, $l=604$, $l=606$, $l=608$, $l=610$, $l=612$, $l=614$, $l=616$, $l=618$, $l=620$, $l=622$, $l=624$, $l=626$, $l=628$, $l=630$, $l=632$, $l=634$, $l=636$, $l=638$, $l=640$, $l=642$, $l=644$, $l=646$, $l=648$, $l=650$, $l=652$, $l=654$, $l=656$, $l=658$, $l=660$, $l=662$, $l=664$, $l=666$, $l=668$, $l=670$, $l=672$, $l=674$, $l=676$, $l=678$, $l=680$, $l=682$, $l=684$, $l=686$, $l=688$, $l=690$, $l=692$, $l=694$, $l=696$, $l=698$, $l=700$, $l=702$, $l=704$, $l=706$, $l=708$, $l=710$, $l=712$, $l=714$, $l=716$, $l=718$, $l=720$, $l=722$, $l=724$, $l=726$, $l=728$, $l=730$, $l=732$, $l=734$, $l=736$, $l=738$, $l=740$, $l=742$, $l=744$, $l=746$, $l=748$,

{ 2, 10, 6, 1, 0, 1, 3, 0, 0, 0, 0, 0, 0, 0, 0 }
 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
 ↑
 7 → Ans = 7

$\text{for}(l=1; \underline{1}; \underline{l=2i})$ $\Rightarrow$ Loop with over
 $\text{if}(a(i) == \infty \parallel a(i) = \infty)$ $\Rightarrow$ Info. bug
 $\{$
 $\quad m = i$
 $\quad \text{return } n;$
 $\}$

$n := 0$

Approx = 2

$$\lambda = 0, \quad \lambda = 1$$

①

find n $l = 2xi$

$\Rightarrow O(\log n)$

②

Same level prob
or
 $\log n + \log n = 2 \log n$

all about Binary

Search